

CHE 321 – Chemical Reaction Engineering Spring 2019

Home Work – 3 Solutions

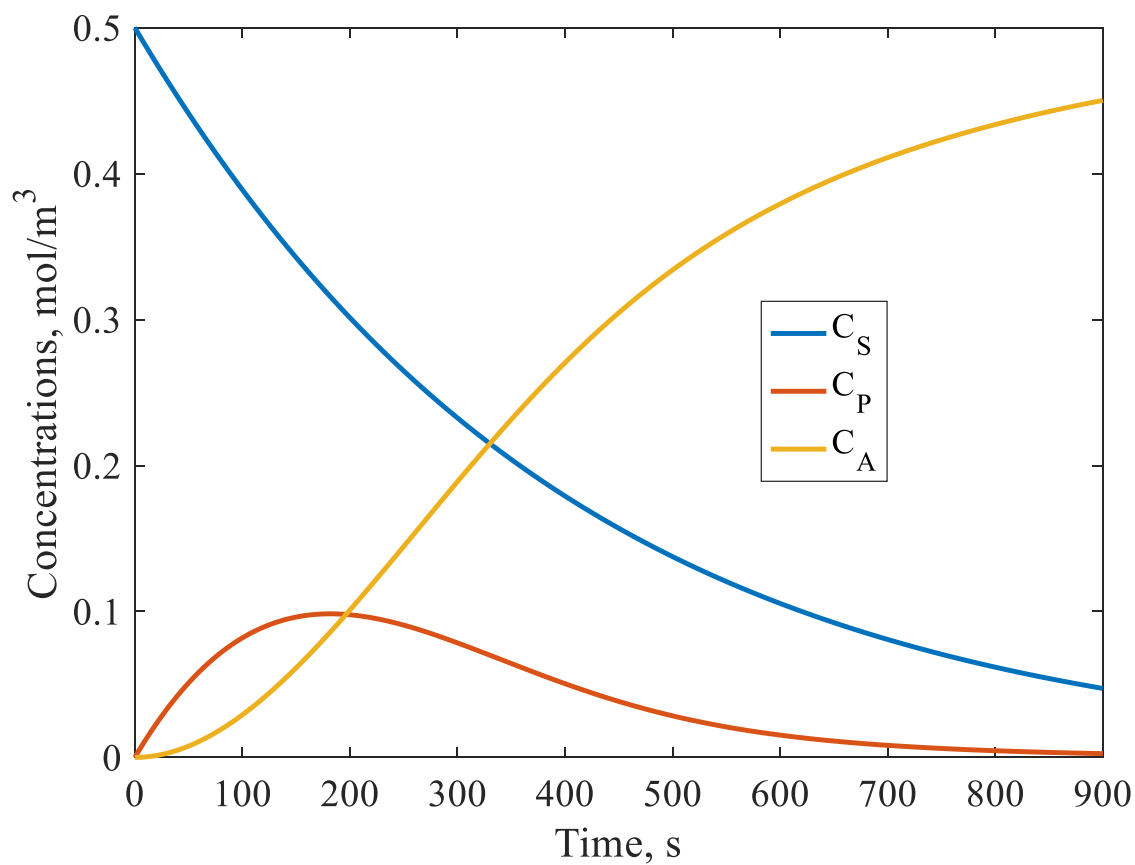
1) Batch Reactor

Batch reactor model equation

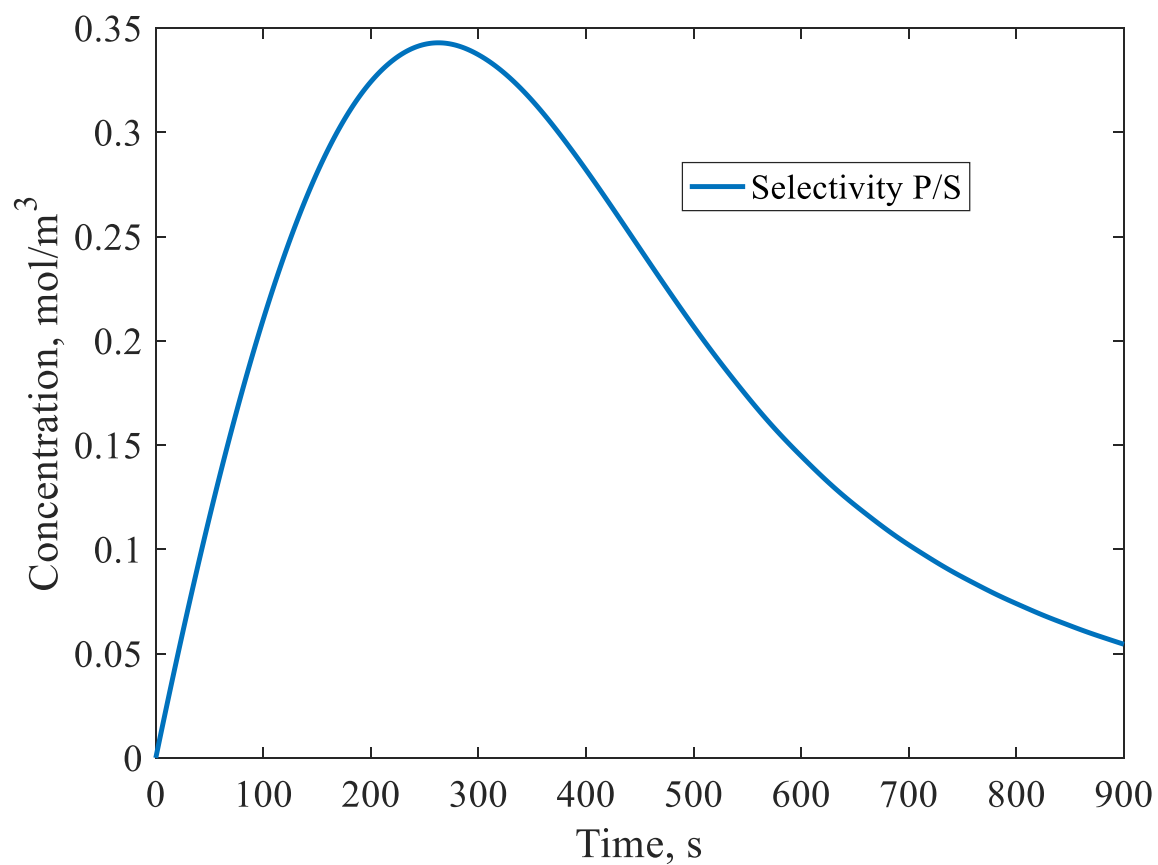
$$\frac{d\underline{C}}{dt} = \underline{\alpha}^T \underline{r} \quad 1$$

$$\frac{d\xi}{dt} = \underline{r} \quad 2$$

$$C_j = C_{j0} + \sum_i \alpha_{ij} \xi_i \quad 3$$

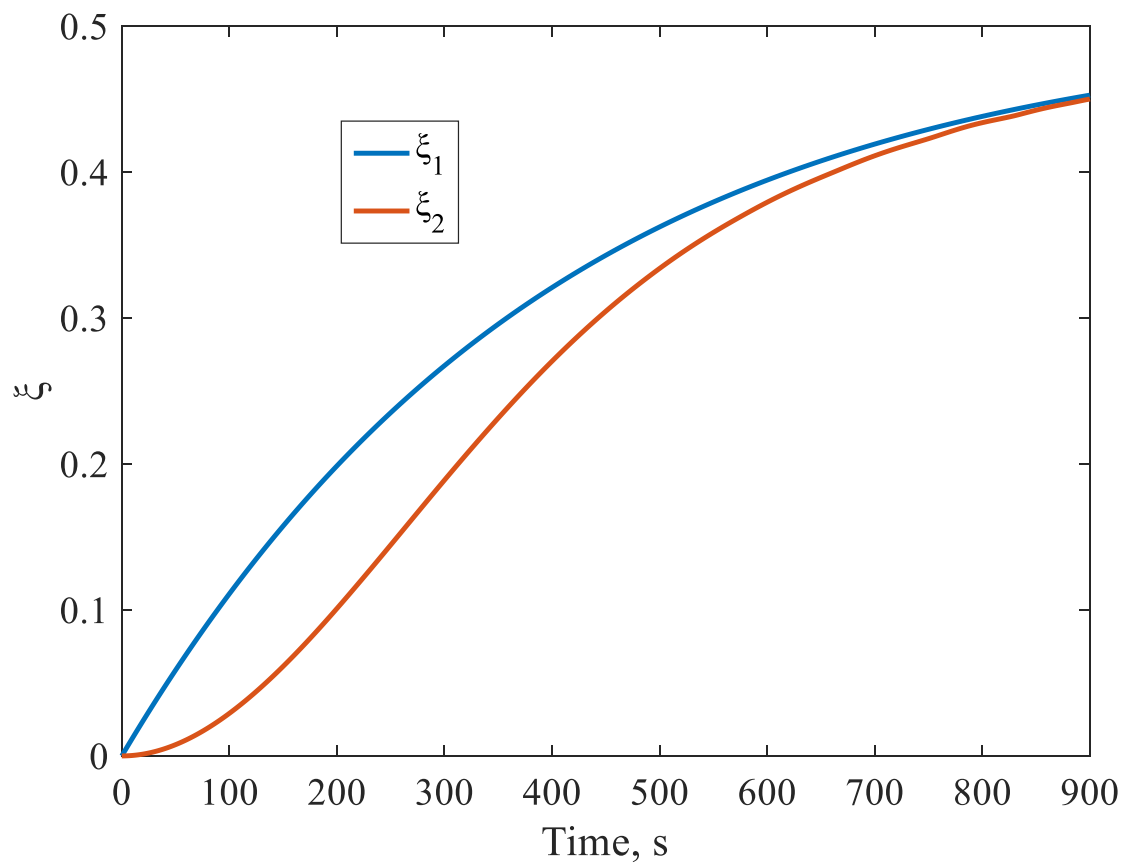


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Selectivity is maximum at around **260 s** and it is the desired operating time.

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2) Plug Flow Reactor

Assumptions

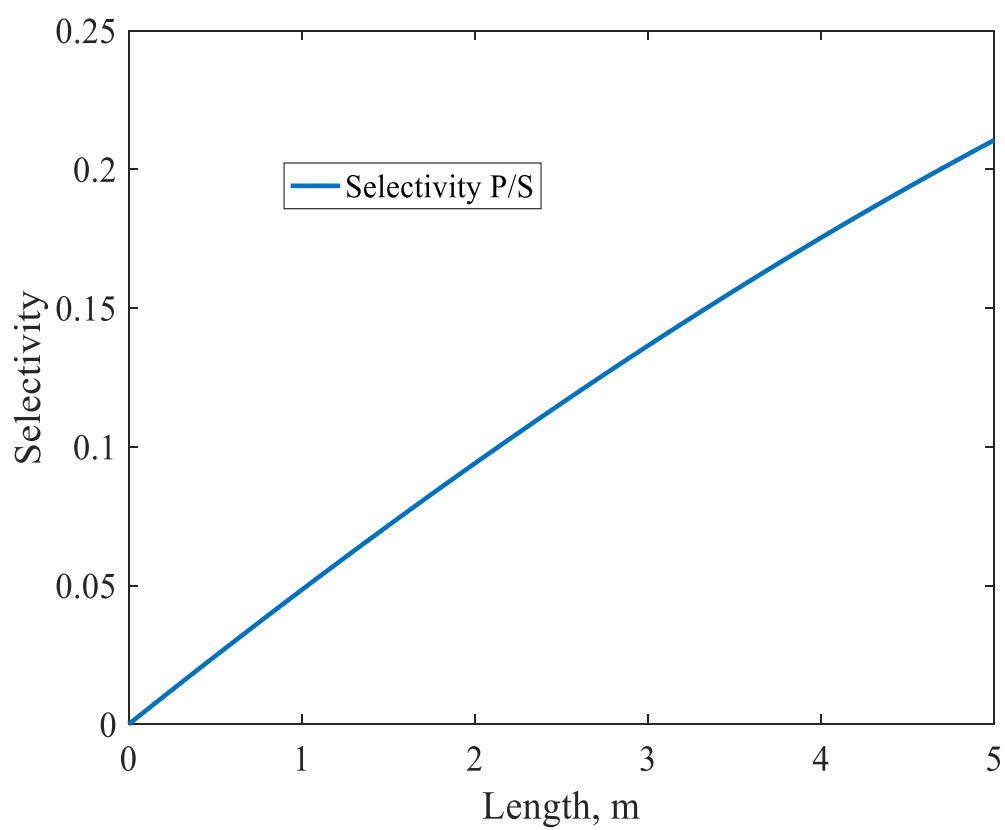
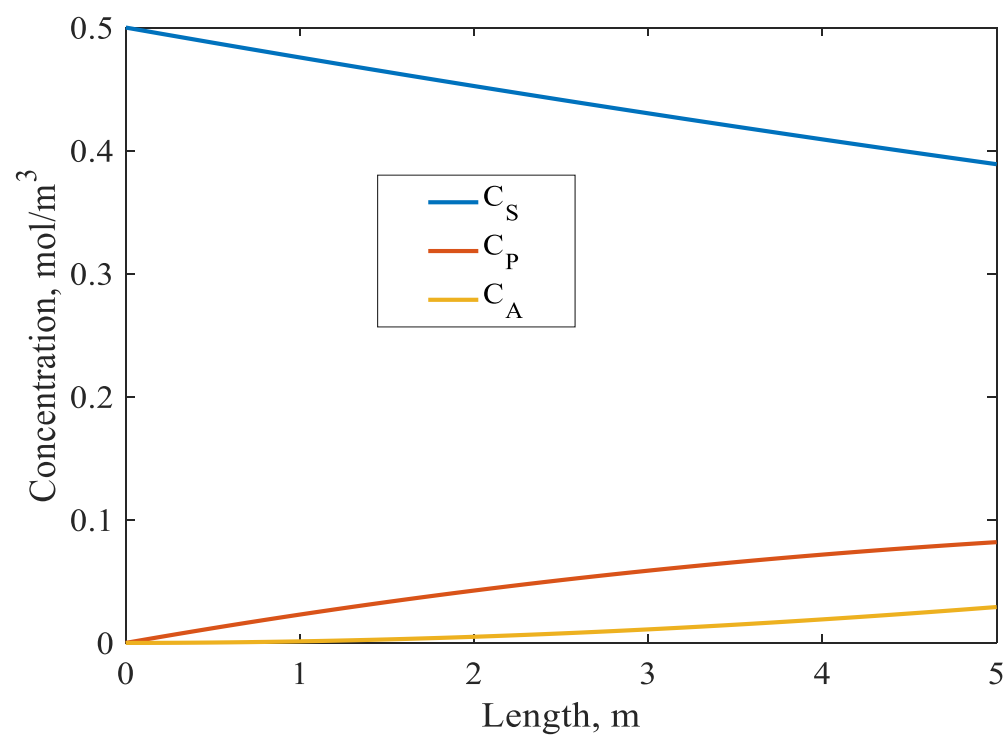
- 1) Steady state
- 2) 1D Cartesian co-ordinates
- 3) Diffusion is negligible
- 4) Constant velocity

$$v \frac{dC}{dz} = \alpha^T r$$

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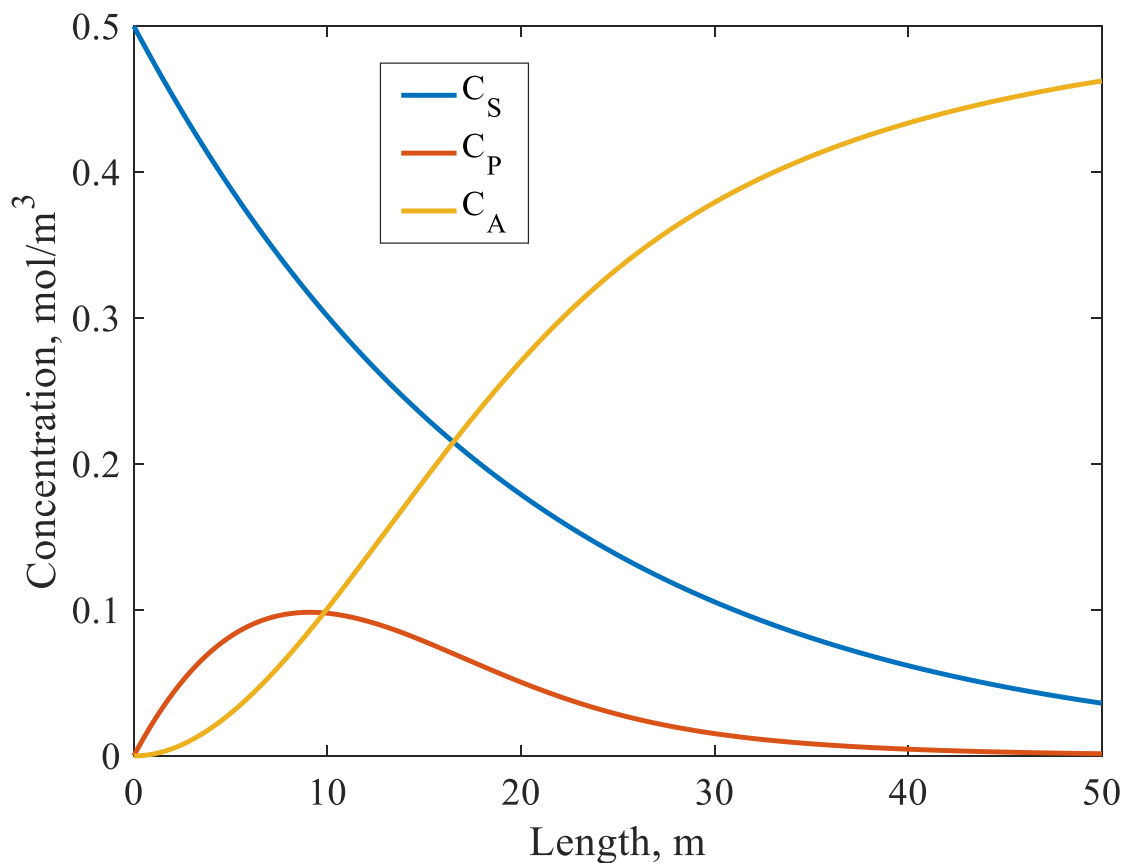
To improve the conversion of the reactant,

- 1) The length of the reactor must be increased.
- 2) The velocity of the reactants should be decreased.

PFR is preferred as the same conversion can be obtained at smaller volume.

Batch operation – 260 s operation in 1 m³ volume of the reactor.

PFR operation – continuous operation in a reactor of length 8m and diameter 10 cm.



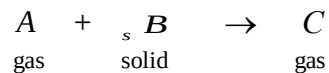
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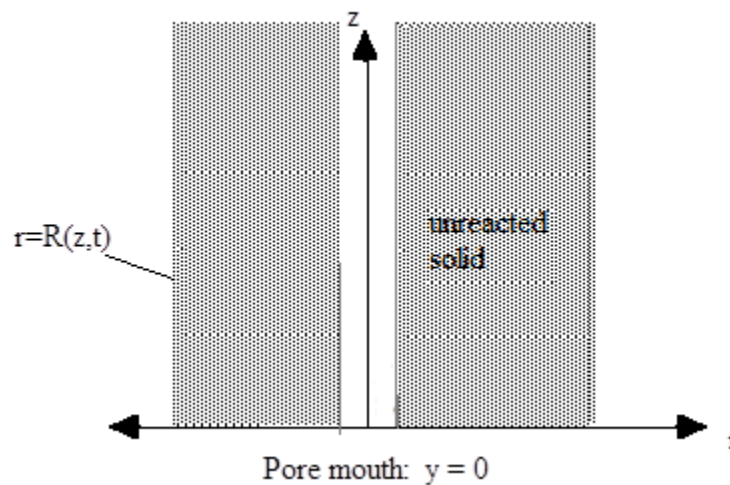
Homework 2 - Solutions

Part A)

Consider a non-catalytic solid-fluid chemical reaction within a **symmetric** pore of **infinite length** and **circular cross section of radius initially** R_0 occurring at the surface and **the solid is in excess** and after some time the system reaches **steady-state in terms of the concentration..** The consumption of the porous solid is to be modeled as via the **analysis of a single pore**. The pore at time t appears as shown in the figure below. The **specific volume is constant**. The **axial diffusion can be neglected**. The System is 2-D (r and z)



The reaction is first order in the concentration of A with an **area-specific rate constant** $\hat{k}$. The concentration of A at the **pore mouth** may be assumed to remain constant at C_{Af} .



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- (i) *Formulate the differential equation for the concentration of gas A neglecting radial diffusion but of course including axial diffusion with coefficient D along the pore. [5 points]*

$$\frac{\partial C_A}{\partial t} + V \cdot \left[\frac{1}{r} \frac{\partial(rC_A)}{\partial r} + \frac{\partial C_A}{\partial z} \right] - \frac{1}{r} \frac{\partial}{\partial r} \left(rD \frac{\partial C_A}{\partial r} \right) = 0$$

- (ii) *Identify the boundary conditions for the diffusion of the gas within the deforming pore. Neglect external mass transfer resistance. State your assumptions. [5 points]*

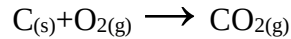
$$\begin{aligned} z = 0, C_A &= C_{Af} \\ r = 0, \frac{\partial C_A}{\partial r} &= 0 \\ r = R, K(C_A - C_{As}) &= K_A C_{As} \end{aligned}$$

- (iii) *Obtain the differential equation for r(z,t) representing the shape of the pore assuming C_s to be the molar density of the solid. Provide initial and boundary conditions for (z,t). [5 points]*

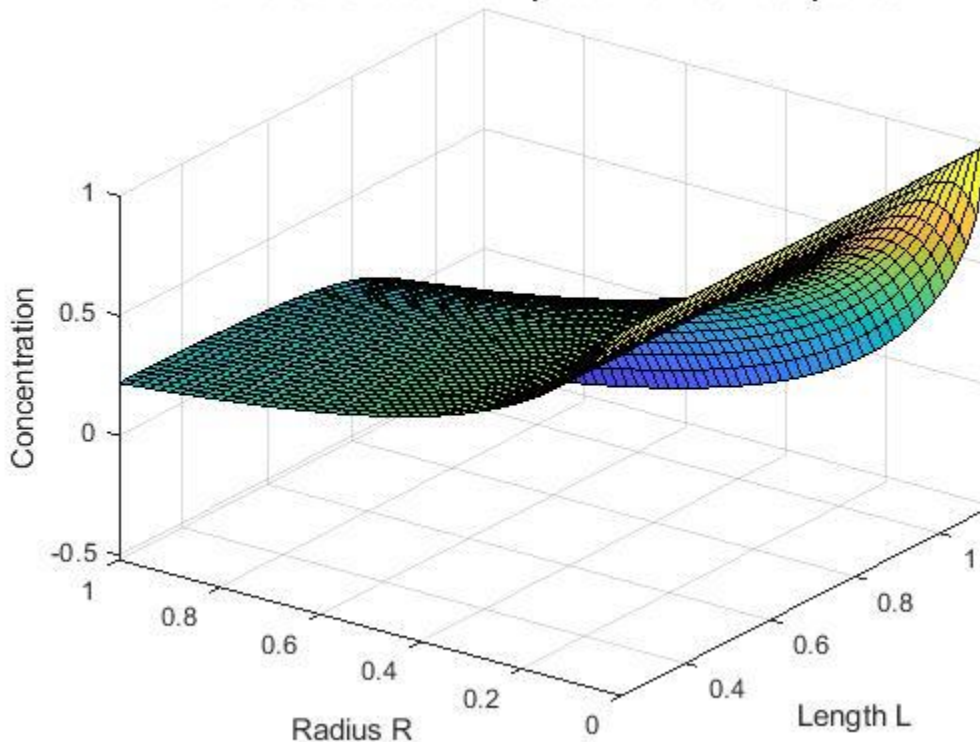
$$\begin{aligned} f(r, z, t) &= 2\pi r z \\ \frac{\partial f}{\partial t} &= \alpha \hat{r} [\nabla \cdot f] \\ 2\pi z \frac{\partial r}{\partial t} &= \alpha \hat{r} \left[\frac{1}{r} \frac{\partial(rf)}{\partial r} \right] \\ 2\pi z \frac{\partial r}{\partial t} &= \alpha \hat{r} \left[\frac{1}{r} \frac{\partial(rf)}{\partial r} \right] \\ 2\pi z \frac{\partial r}{\partial t} &= \alpha \hat{r} [4\pi z] \\ \frac{\partial r}{\partial t} &= 2\alpha \hat{r} \\ r &= 2\alpha K C_{As} t + C \\ t = 0, r &= r_0 \\ r &= 2\alpha K C_{As} t + r_0 \end{aligned}$$

Part B)

Consider the combustion of the coal. The air flows at velocity of 1 cm/s.. the diffusivity coefficient of air is $1.6 \times 10^{-5} \text{ m}^2/\text{s}$ and $C_{A,f}$ is 1 mol/m^3 . The concentration of the oxygen is 0.2 mol/m^3 . The reaction constant is $0.00009 \text{ (cm}^2.\text{s)}^{-1}$ at 400°C . And mass transfer coefficient of air is $1.5 \times 10^{-3} \text{ cm/s}$. The pore diameter at $t=0$ is 0.5 cm. and the volume of $13.3 \text{ cm}^3 \text{ coal/mol O}_2$ is consumed by air. plot the profile of reaction rate of the reaction as function of radius and length in 30 minutes of combustion at length of 1 cm . [5 points]



Numerical solution computed with 20 mesh points.



```
function r=Homework4
clc
V=1; %cm/s;
r0=0.25; %cm;
DAB=1.6*10^-5*10^4; %Cm^2/s;
CAF=1; %mol/s
CAS=0.2; %mol/s
Kr=0.00009; %1/(Cm^2.s)
alphap=13.3; %cm^3/mol ...1mol O2* (1mol C/1mol O2)* (12 gr C/1molC) *
(10^3 cm^3/900 gr)
```



```

Kc=1.5*10^(-3); %cm/s
t1=30*60;% s
t=linspace(0,t1,50);
%r=2??CAst+r_0
r=2*alphap*Kr*CAS*t+r0;
% (??C_A)/?t+V.[1/r  (??(rC?_A))/?r+(?C_A)/?z]-1/r  ?/?r  (rD
(?C_A)/?r)=0
% z=0,C_A=C_Af
% r=0,(?C_A)/?r=0
% r=R,(?C_A-C_(A,))=?AC_As

m = 1;
z = linspace(0,1,50);
R= r;

sol = pdepe(m,@pdex1pde,@pdex1ic,@pdex1bc,z,r);
% Extract the first solution component as z.
ca = sol(:, :, 1);

% A surface plot is often a good way to study a solution.
figure;
surf(z,r,ca)

title('Numerical solution computed with 20 mesh points.')
xlabel('Length L')
ylabel('Radius R')

end

```